

Digital Signal Processing 1

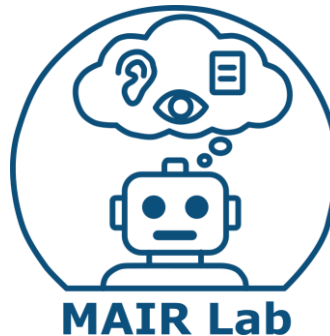
안인규 (Inkyu An)

Speech And Audio Recognition
(오디오 음성인식)

<https://mairlab.kookmin.ac.kr/>

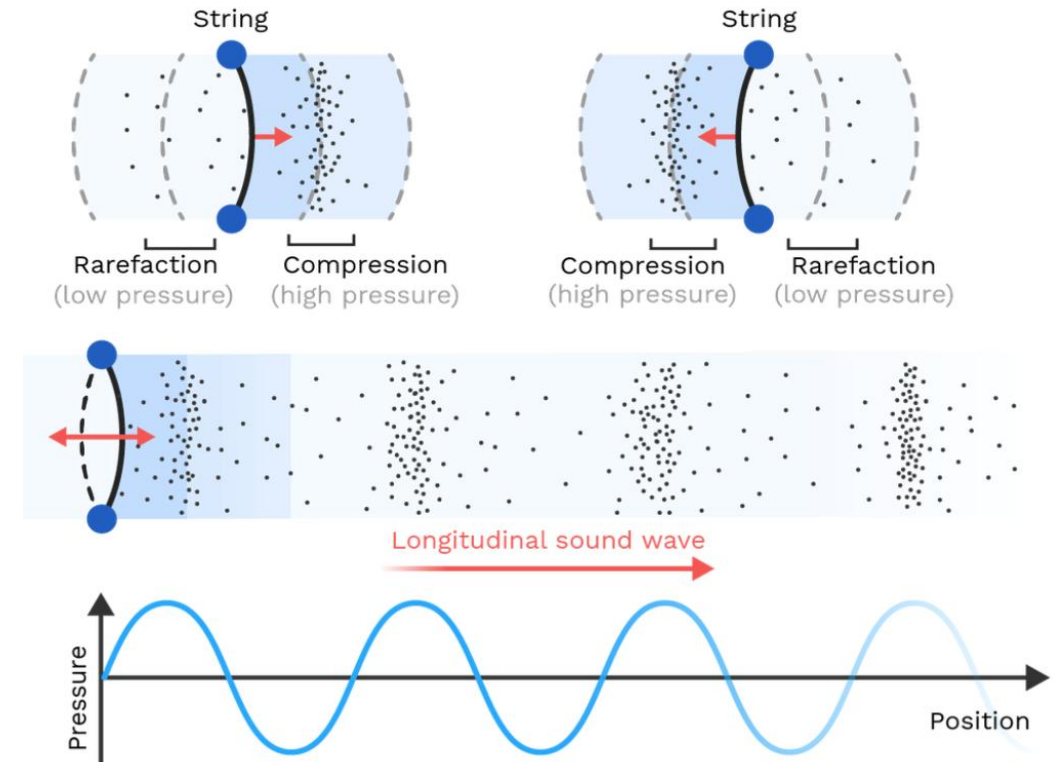


This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
<https://github.com/markovka17/dla>



What is Sound?

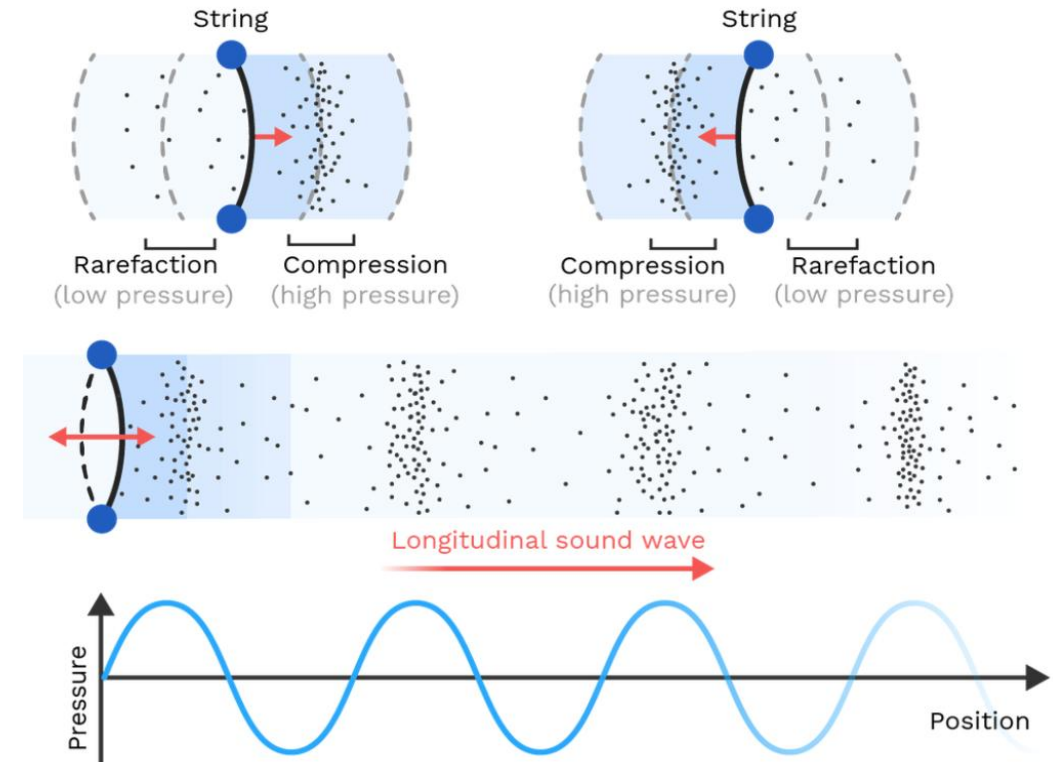
- The air around us is filled with molecules.
- When you pull a guitar string, it creates a vibration that moves through the molecules in the air.
- The regions of high pressure are compressions and the regions of low pressure as rarefactions.



What is Sound?

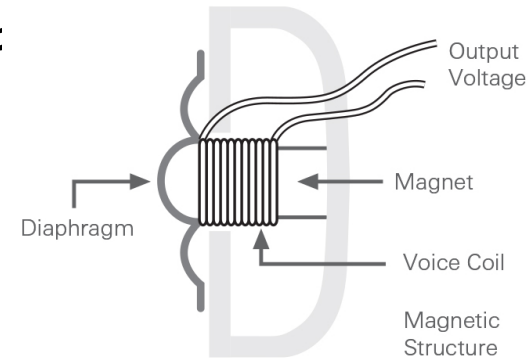
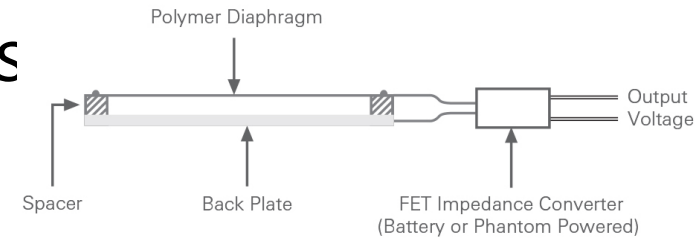
The microphone detects these variations in pressure.

When we plot pressure, **relative to atmospheric** pressure, against the position near the string, we see the familiar sinusoidal waveform.

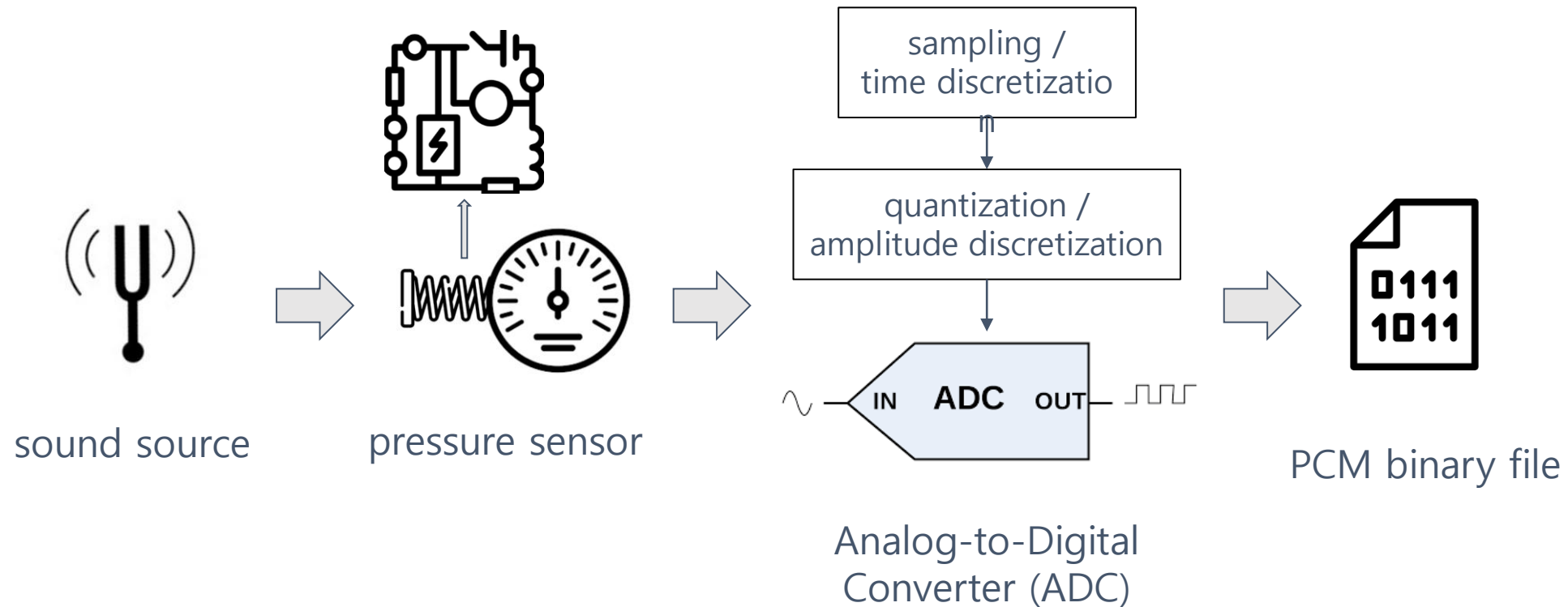


Recording

- To record people use microphones
- Microphone picks up these air oscillations in continuous form
- These oscillations are converted into an analog signal and then a digital signal

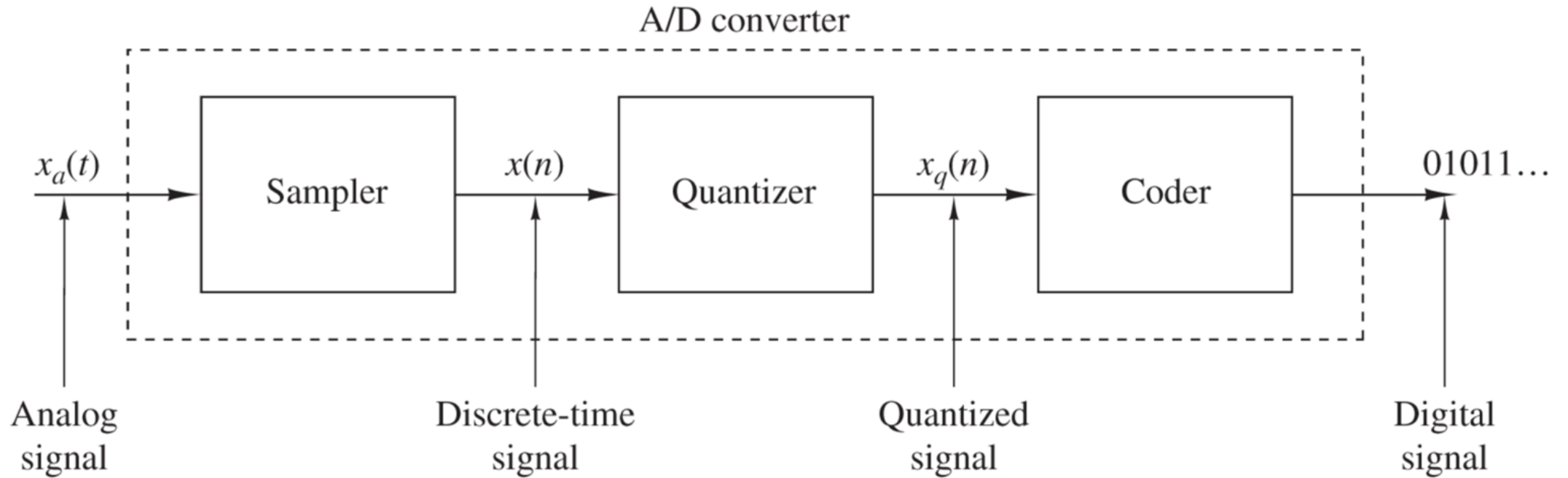


Analog and Digital Signals



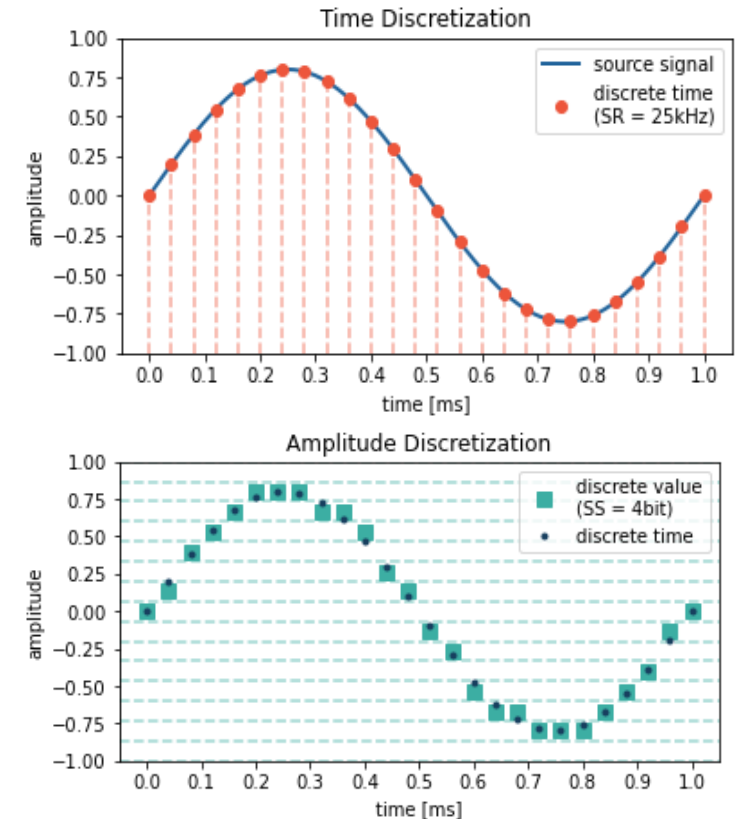
The sinusoidal waveform follows from physics equations, see [this slide](https://theory.labster.com/sound-waves-dbs/)
<https://theory.labster.com/sound-waves-dbs/>

Analog and Digital Signals



Waveform as Pulse-Code Modulation (PCM)

- **Time discretization:** we represent the analog signal as a sequence of samples measured at discrete points in time
 - **Sample rate** – number of audio samples per second (8kHz, 22.05kHz, 44.1kHz)
- **Amplitude discretization:** round continuous amplitude to the nearest discrete value
 - **Bit depth** – number of bits per sample (eg. 8, 16, 24, 32 bits)
 - **Bit rate** = bit-depth * sample-rate * audio-channels
- **Number of channels:** number of signals recorded in parallel (e.g., mono vs. stereo)



Properties of Waveforms: Intensity

Intensity is defined to be the *energy* (E) per unit *area* (A) and time unit (t) carried by a wave:

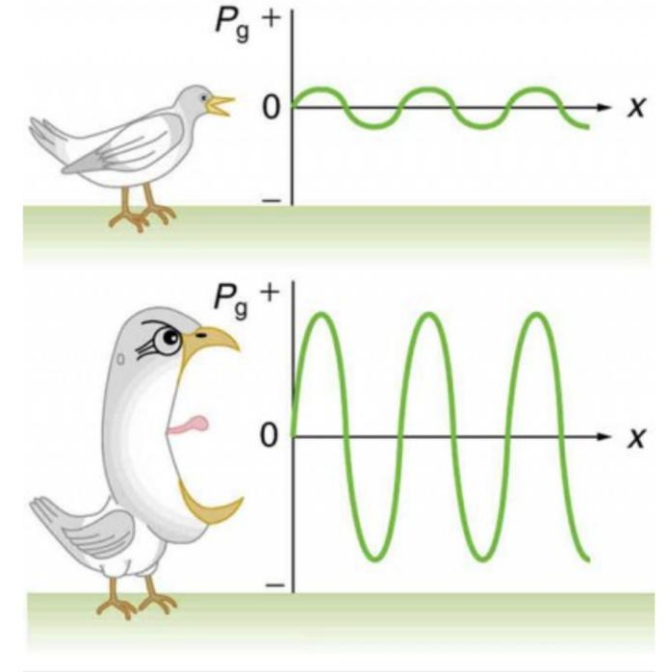
$$I = \frac{E}{tA} = \frac{P}{A} \quad \leftarrow \text{power}$$

The intensity of a sound wave is proportional to its amplitude squared:

$$I \sim (\Delta p)^2$$

For a discrete-time signal $\{p_n\}_{1:T}$ of length T :

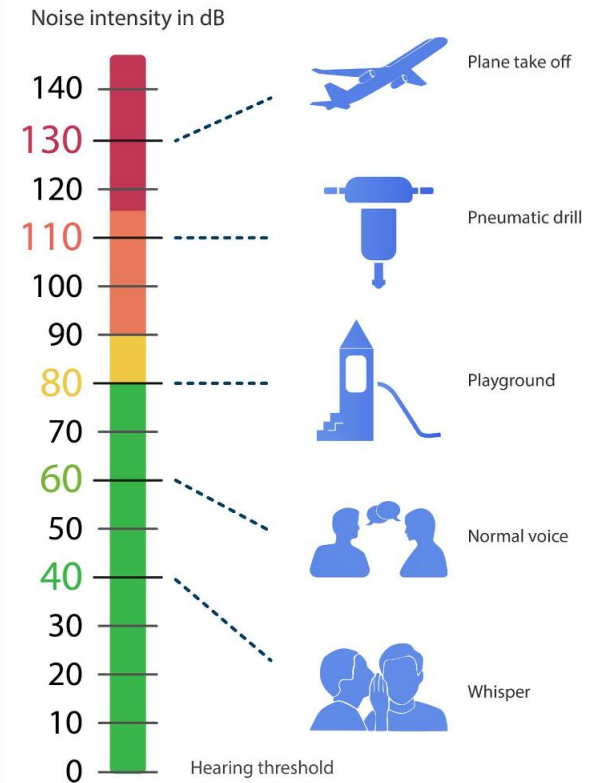
$$I_n \sim p_n^2$$



Properties of Waveforms: Loudness

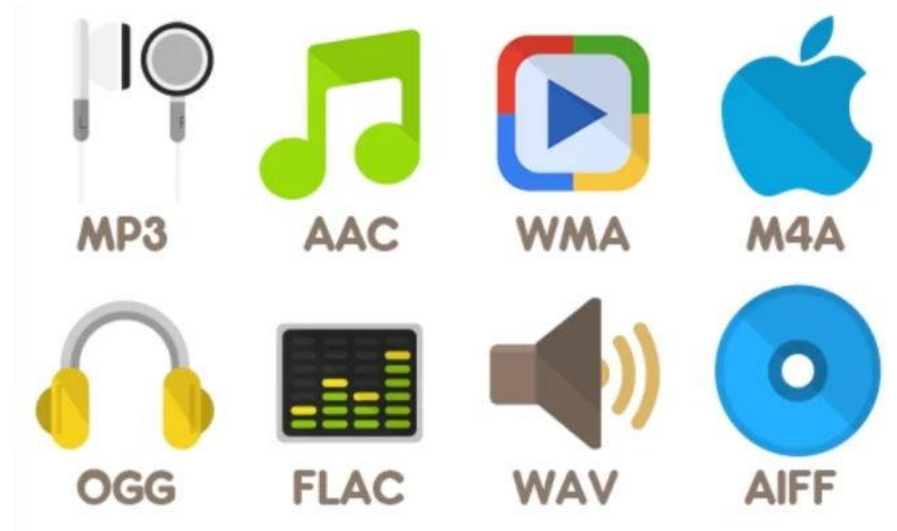
- **Loudness** is intensity measured in decibel
- **bel** reports $\log_{10}$ of ratio between measuring signal and reference signal
- In physics, **decibels** are used instead of bels because 1 bel is too large

$$I_{dB} = 10 \log \left(\frac{I}{I_0} \right) = 10 \log \left(\frac{p^2}{p_0^2} \right) = \\ = 20 \log \left(\frac{p}{p_0} \right)$$



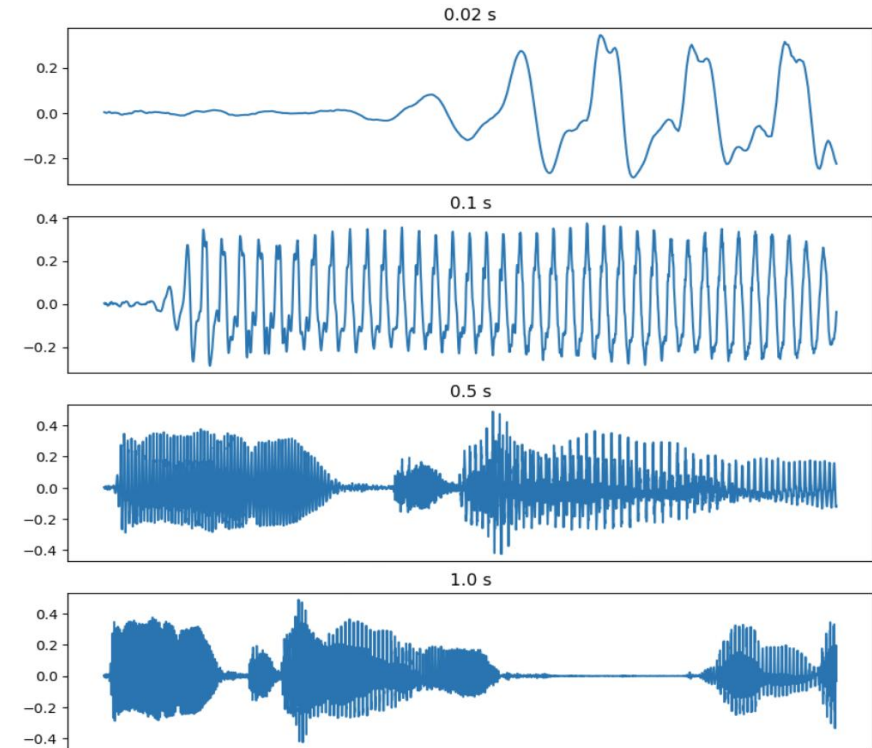
What about audio formats?

- Uncompressed: WAV, AIFF
- Lossless compression: FLAC, ALAC
- Lossy compression: MP3, Opus



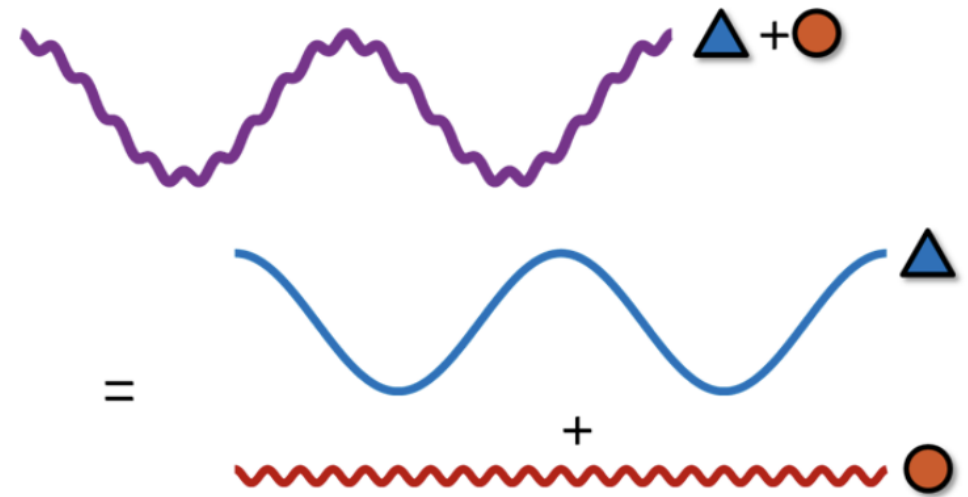
Difficulties in using waveforms

- Waveform is really long (e.g. 44K samples / sec)
- Waveforms provide limited insight into a recording's pitch and speech content
- Can we get a more compact and informative sound representation?



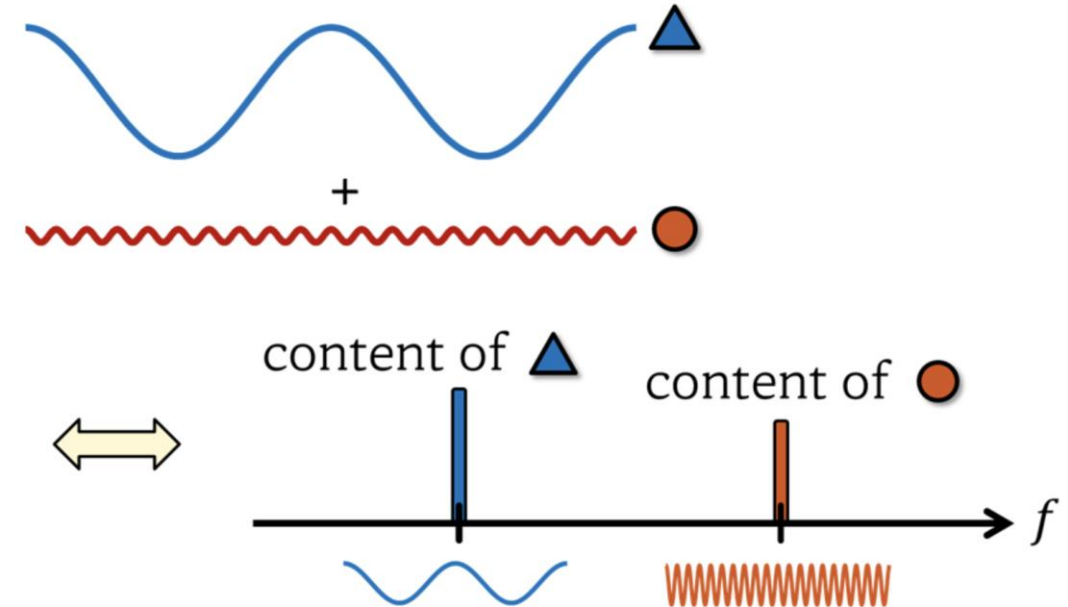
Fourier Analysis

- At its core, Fourier analysis breaks down complex signals into their frequency components.
- It's similar to breaking down a song into its individual musical notes.



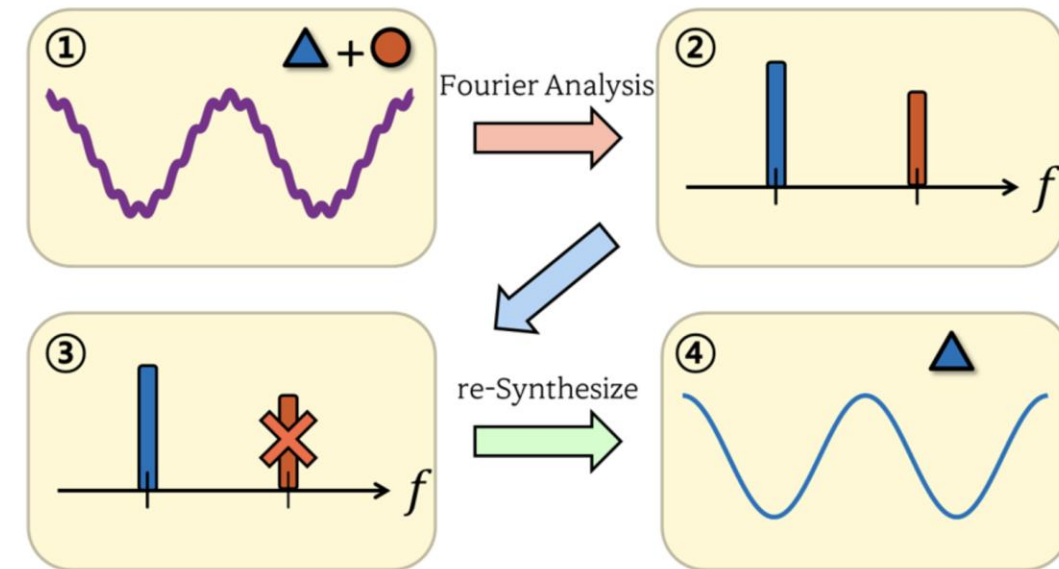
Fourier Analysis

- The amplitudes of these frequency components appear on the frequency axis rather than the time axis, forming what's called the (frequency) spectrum



Fourier Analysis

- The usefulness of Fourier analysis:
 - It converts a long sequence of time samples into a compact frequency representation.
 - It enables analysis of frequency components and filtering of unwanted frequencies.



Fourier Analysis

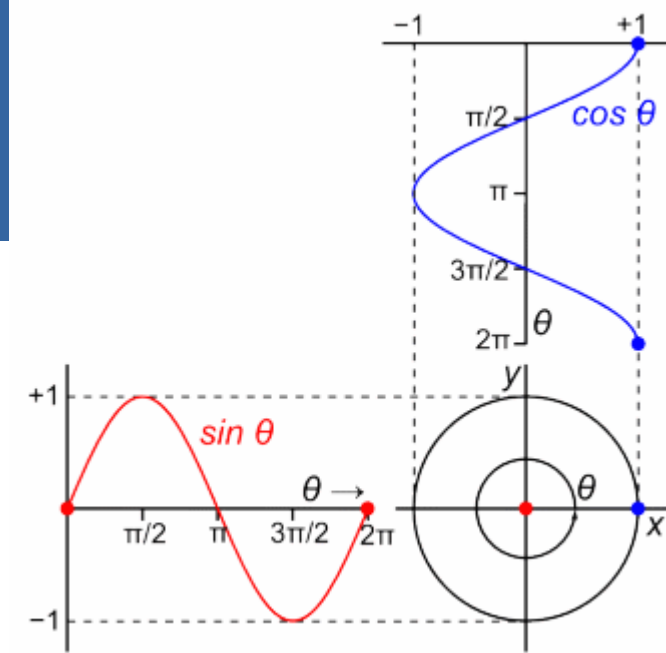
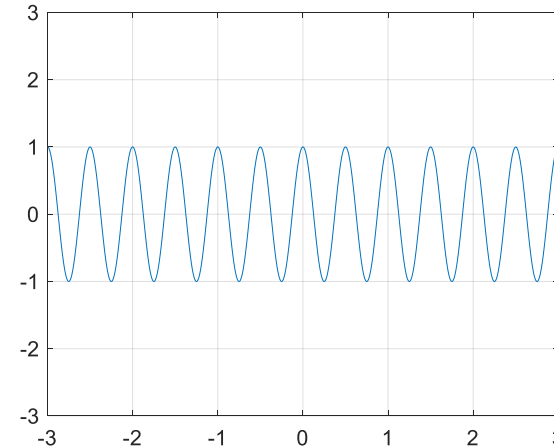
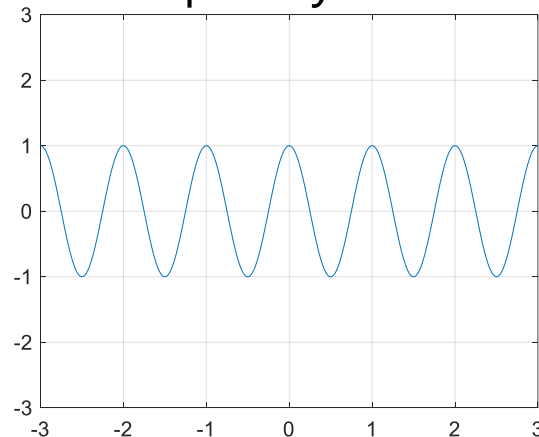
- Sinusoidal Signal

- A signal that oscillates periodically at a specific frequency

- $x(t) = A \cos(2\pi ft + \phi)$

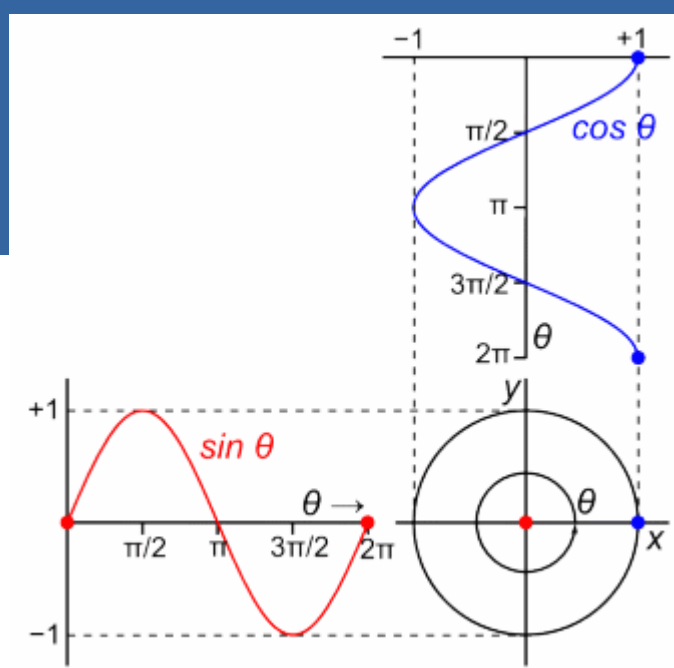
- A : amplitude
- f : frequency [Hz]
- ϕ : Phase [rad]
- t : Time [s]

- What is the frequency of the below graph?



Fourier Analysis

- Sinusoidal Signal
 - A signal that oscillates periodically at a specific frequency
 - $x(t) = A \cos(2\pi f t + \phi)$
 - A : amplitude
 - f : frequency [Hz]
 - ϕ : Phase [rad]
 - t : Time [s]
 - What is the angular frequency?



Fourier Analysis

- Sinusoidal Signal

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- $x(t) = A \cos(2\pi f t + \phi)$

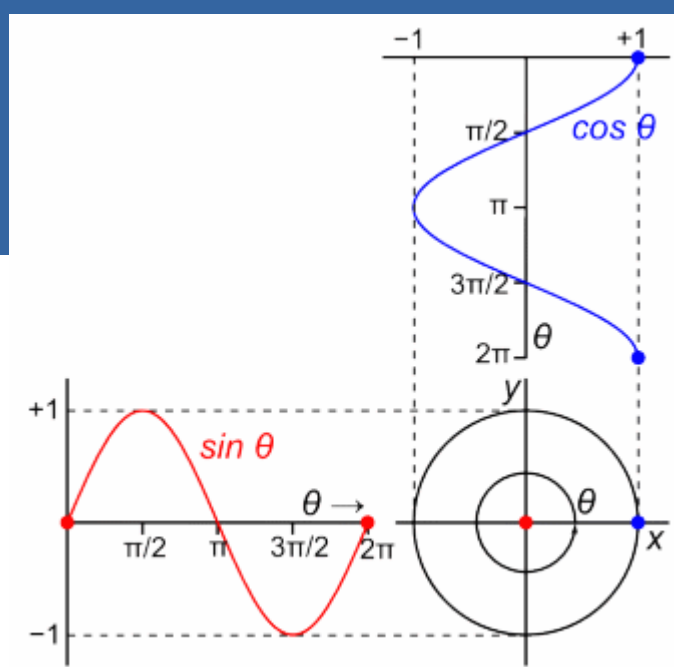
- A : amplitude
- f : frequency [Hz]
- ϕ : Phase [rad]
- t : Time [s]

- What is the angular frequency?

- $\omega = 2\pi f$ [rad/s]
- $x(t) = A \cos(\omega t + \phi)$

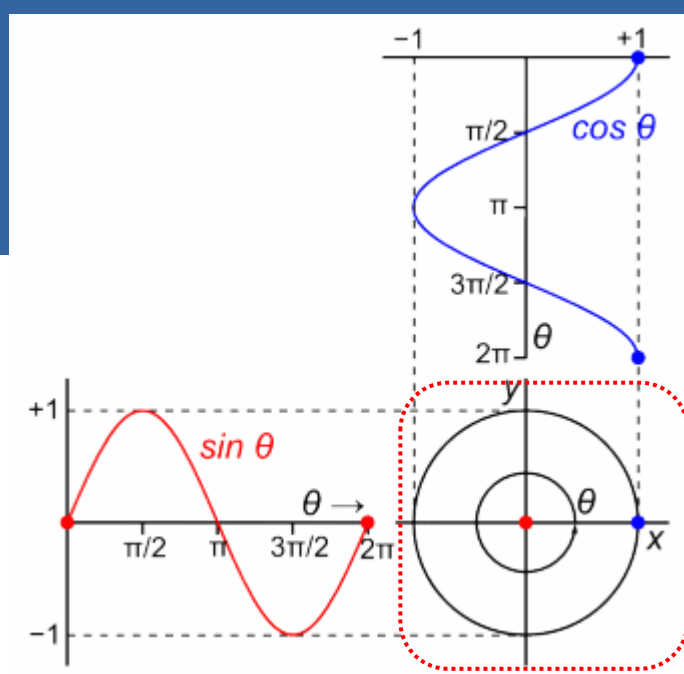
- What is the radian (rad)?

- $1 \text{ rad} = 180/\pi$



Fourier Analysis

- Sinusoidal Signal $\rightarrow$ Euler's formula
 - A representation that combines sine and cosine into a single exponential form
 - $e^{j\theta} = \cos \theta + j \sin \theta$
 - A sine and cosine signals can be viewed as the projection of a rotating complex exponential onto the real axis
 - $\text{Re}\{e^{j\theta}\} = \cos \theta$
 - $\text{Im}\{e^{j\theta}\} = \sin \theta$



Fourier Analysis

- Fourier Series

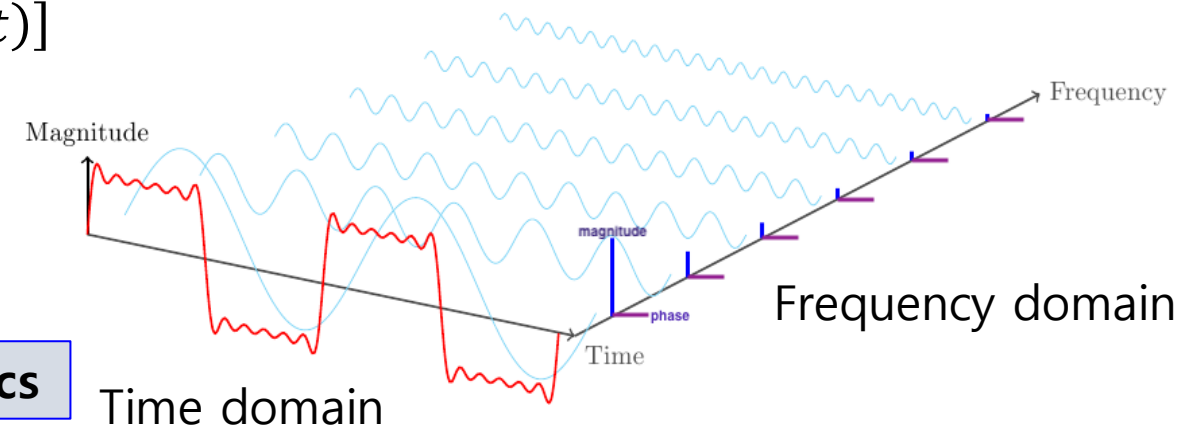
- Can a complex periodic signal be represented as a sum of simple sinusoidal signals? Yes!
 - $x(t + T_0) = x(t)$
 - Fundamental frequency: $f_0 = \frac{1}{T_0}$
 - Angular frequency: $\omega_0 = \frac{2\pi}{T_0}$
 - In Fourier Series, a periodic signal is represented as a sum of sinusoids whose frequencies are integer multiples of the fundamental frequency:
 - $f_0, 2f_0, 3f_0, \dots$
 - These frequency components are called harmonics

Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as

- $$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$
 - $a_0/2$: DC component
 - $k = 1$: fundamental frequency
 - $k = 2$: second harmonics
 - ...

Periodic Signal = DC + Fundamental + Harmonics



Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as **Fourier Series: sine-cosine form**

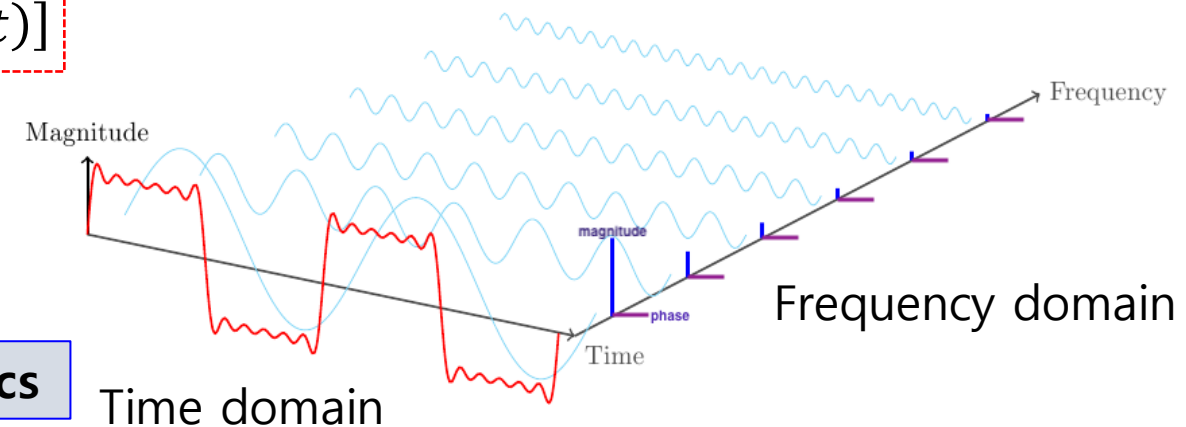
- $$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

- $a_0/2$: DC component
- $k = 1$: fundamental frequency
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- ...

Periodic Signal = DC + Fundamental + Harmonics

Fourier Series: exponential form

- $$x(t) = \sum_{n=-\infty}^{\infty} c_k e^{jn\omega_0 t}$$



Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as **Fourier Series: sine-cosine form**

- $$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

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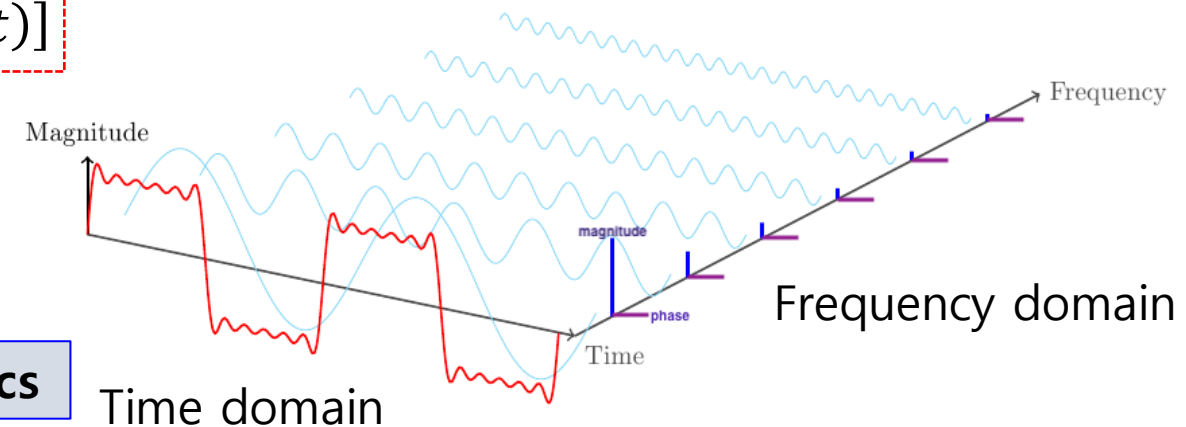
Periodic Signal = DC + Fundamental + Harmonics

Fourier Series: exponential form

- $$x(t) = \sum_{n=-\infty}^{\infty} c_k e^{jn\omega_0 t}$$

- $$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos\left(2\pi \frac{n}{P} t - \phi_n\right)$$

**Fourier Series:
Amplitude-phase form**

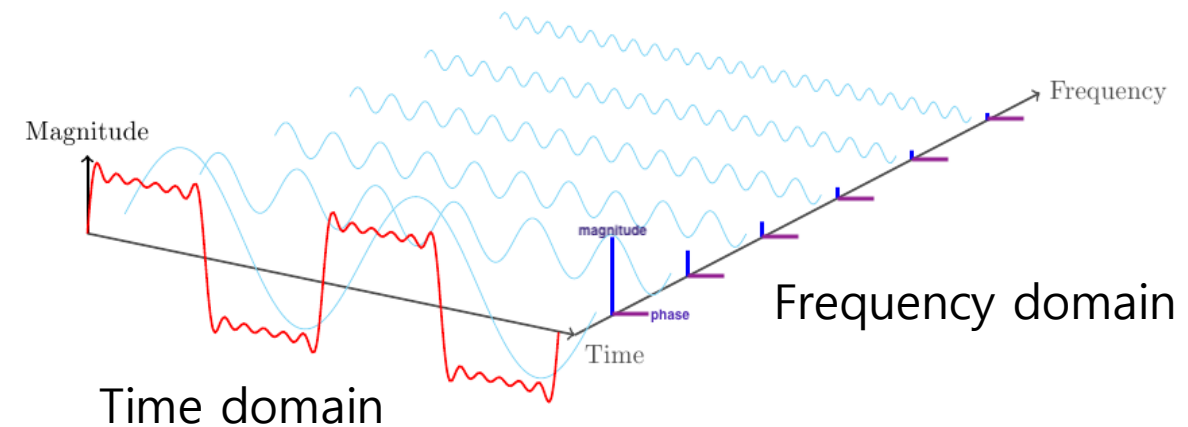


Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as

Fourier Series:
Amplitude-phase form

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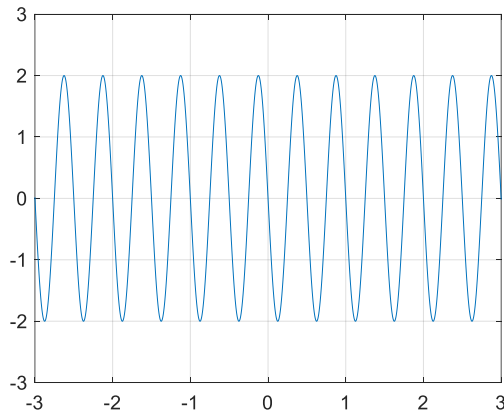


Fourier Series

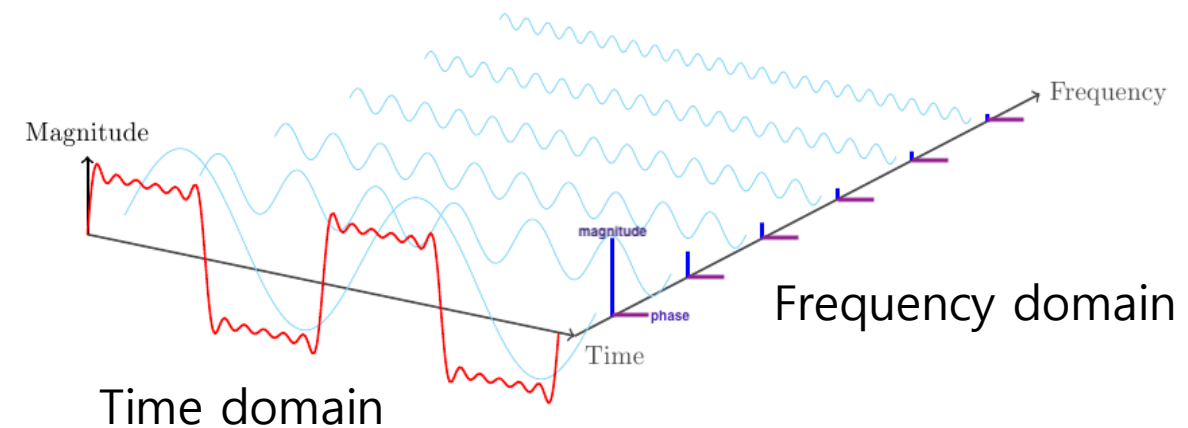
- Any absolutely integrable periodic function $x(t)$ with period P can be represented as

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \overbrace{A_n}^{\text{Amplitude}} \cos\left(2\pi \underbrace{\frac{n}{P}t}_{\text{Frequency}} - \underbrace{\phi_n}_{\text{Phase}}\right)$$

- What is the amplitude and the phase?



- Amplitude?
- Phase?



Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos\left(2\pi \frac{n}{P} t - \phi_n\right)$$

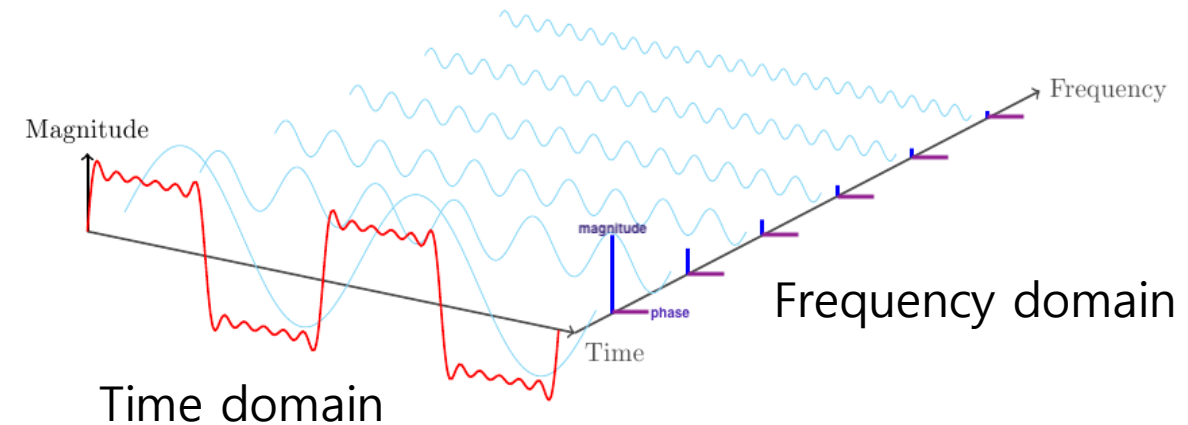
Fourier Series: Amplitude-phase form

$$= a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right)$$

Fourier Series: sine-cosine form

$$= \sum_{n=-\infty}^{\infty} c_n e^{2\pi i \frac{n}{P} t}$$

Fourier Series: exponential form



Fourier Series: Exponential form

- Fourier Series & Fourier Coefficient
 - Fourier Series:** exponential form

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i \frac{n}{P} t}$$

Frequency component
(Frequency basis)

pure frequency component at $n\omega_0$

$n=0 \rightarrow 0$ Hz (DC)

$n=1 \rightarrow f_0$

$n=2 \rightarrow 2f_0$

...

- $\omega_0 = \frac{2\pi}{P}$

- Frequency basis:

$$\dots, e^{-j2\omega_0 t}, e^{-j\omega_0 t}, 1, e^{j\omega_0 t}, e^{j2\omega_0 t}, \dots$$

- What is the basis?

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$e_i^T e_j = 0 \quad (i \neq j)$$

Orthogonal basis

- Frequency basis:

Inner product

$$\phi_n(t) = e^{jn\omega_0 t}$$

$$\langle \phi_n, \phi_m \rangle = \int_0^P \phi_n(t) \phi_m^*(t) dt$$

$$= \int_0^P e^{jn\omega_0 t} e^{jm\omega_0 t} dt = \int_0^P e^{j(n-m)\omega_0 t} dt$$

$$\langle \phi_n, \phi_m \rangle = \begin{cases} P, & n \neq m \\ 0, & n = m \end{cases}$$

Fourier Series: Exponential form

- Fourier Series & Fourier Coefficient
 - **Fourier Series:** exponential form

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i \frac{n}{P} t}$$

Fourier coefficient

$$c_n = \frac{1}{P} \int_{-\frac{P}{2}}^{\frac{P}{2}} x(t) e^{-2\pi i \frac{n}{P} t} dt$$

$$\phi_k(t) = e^{jk\omega_0 t}$$

- How do we extract only a specific c_k from a given $x(t)$
- To obtain the coefficient of a vector along a particular basis direction, we take the inner product with that basis vector.
- $\langle x, \phi_k \rangle = \int_0^P x(t) \phi_k^*(t) dt = \int_0^P x(t) e^{-jk\omega_0 t} dt$

전개하면,



Fourier Series

- Any absolutely integrable periodic function $x(t)$ with period P can be represented as

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(\omega_0 n t - \phi_n)$$

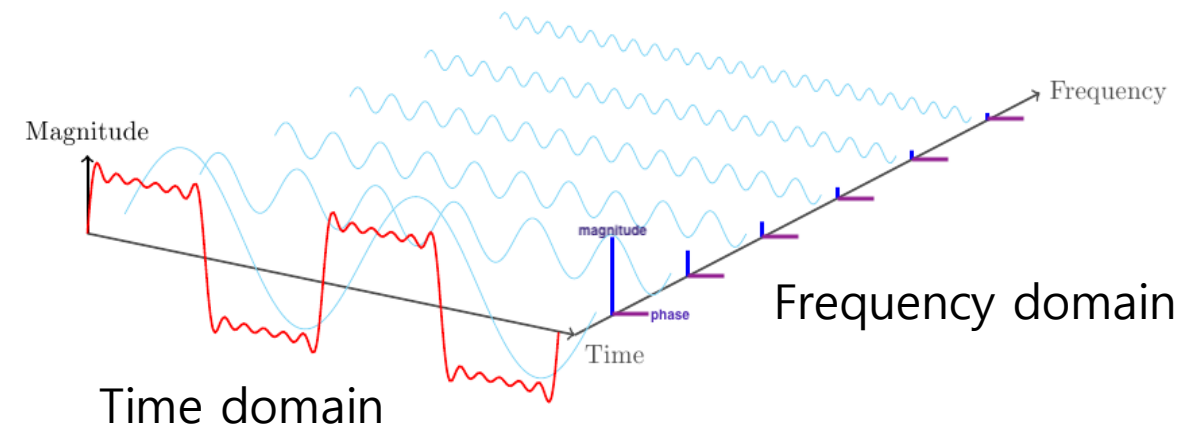
Fourier Series: Amplitude-phase form

$$= a_0 + \sum_{n=1}^{\infty} (a_n \cos(\omega_0 n t) + b_n \sin(\omega_0 n t))$$

Fourier Series: sine-cosine form

$$= \sum_{n=-\infty}^{\infty} c_n e^{j\omega_0 n t}$$

Fourier Series: exponential form



Fourier Transform

- Fourier Series & Fourier Coefficient

P : the period

$$c_n = \frac{1}{P} \int_{-P/2}^{P/2} x(t) e^{-2\pi i \frac{n}{P} t} dt$$

Fourier coefficient

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\omega_0 n t}$$

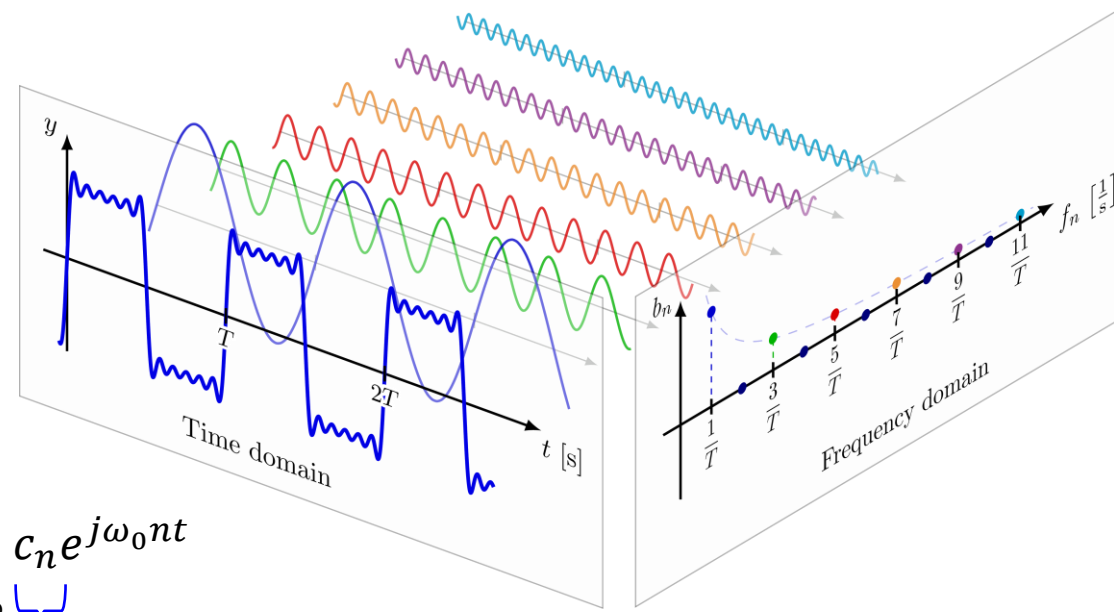
$$\omega_0 = \frac{2\pi}{P}$$

$$\frac{1}{P} = \frac{\Delta\omega}{2\pi}$$

Let's assume $P \rightarrow \infty \rightarrow \omega_0 \rightarrow 0 \rightarrow \Delta\omega = \omega_0 \rightarrow 0$

- Fourier Transform!

- A mathematical formula that allows us to decompose any signal into its individual **frequencies** and the frequency's **amplitude**



$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Frequency

Original signal

Fourier transform

Fourier Transform

- Fourier Series & Fourier Coefficient

P : the period

$$c_n = \frac{1}{P} \int_{-P/2}^{P/2} x(t) e^{-j\omega_0 n t} dt$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\omega_0 n t}$$

Fourier Series

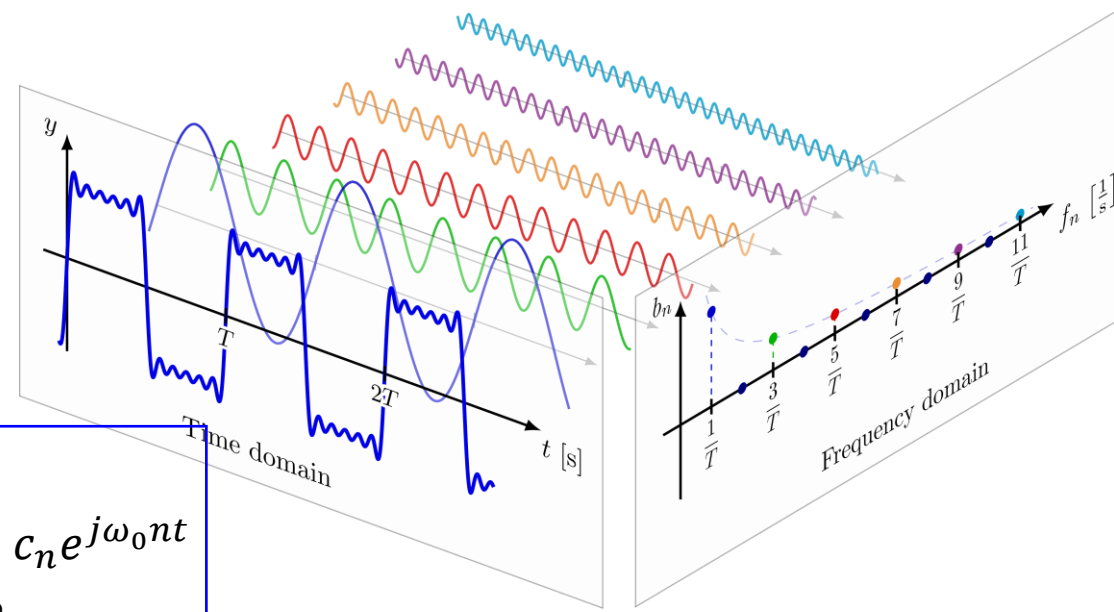
$$\omega_0 = \frac{2\pi}{P}$$

$$\frac{1}{P} = \frac{\Delta\omega}{2\pi}$$

- Let's assume $P \rightarrow \infty \rightarrow \omega_0 \rightarrow 0 \rightarrow \Delta\omega = \omega_0 \rightarrow 0$
 $\Sigma \rightarrow \int$

- Fourier Transform!

- A mathematical formula that allows us to decompose any signal into its individual **frequencies** and the frequency's **amplitude**



Original
signal

Frequency

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Fourier transform

Fourier Transform



- Fourier Transform!
 - A mathematical formular that allows us to decompose any signal into its individual **frequencies** and the frequency's **amplitude**

Frequency

Original signal

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-2\pi i f t} dt$$

Fourier transform

- There is also the inverse Fourier Transform:

Original signal

Frequency

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{2\pi i f t} dt$$

Inverse Fourier transform

Fourier Series → DTFS

- Fourier Series & Fourier Coefficient

Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i \frac{n}{P} t}$$

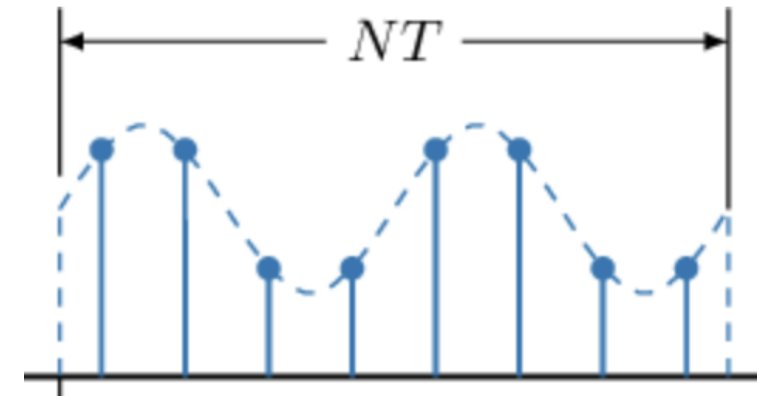
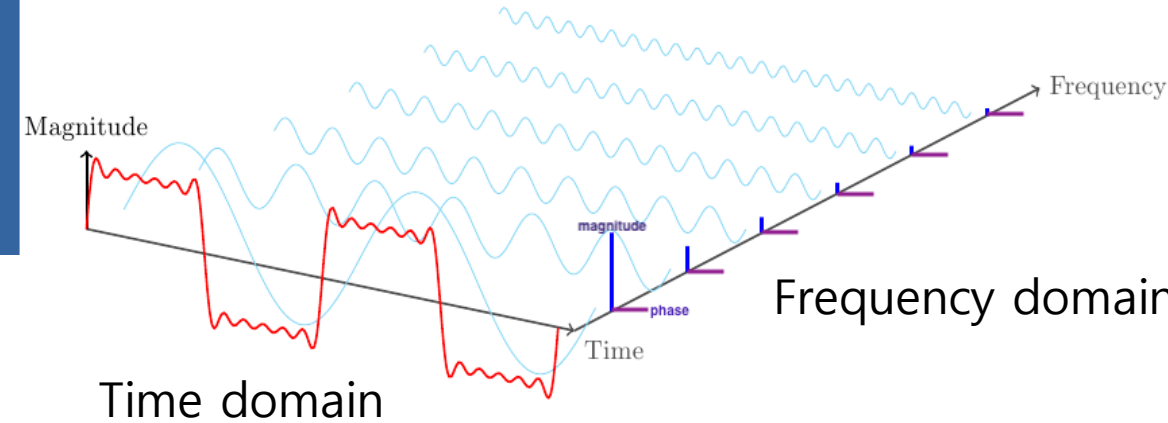


Fourier coefficient

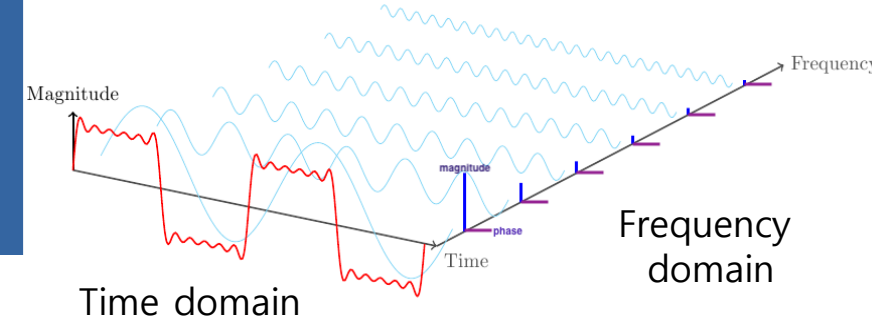
$$c_n = \frac{1}{P} \int_{-P/2}^{P/2} x(t) e^{-2\pi i \frac{n}{P} t} dt, \quad \phi_k(t) = e^{jk\omega_0 t}$$

- Discrete-Time Fourier Series

- In practice our time signal is time-limited and contains N non-zero samples taken with a sampling period T seconds.
- Let's explore what happens when we try to find its Fourier coefficients.

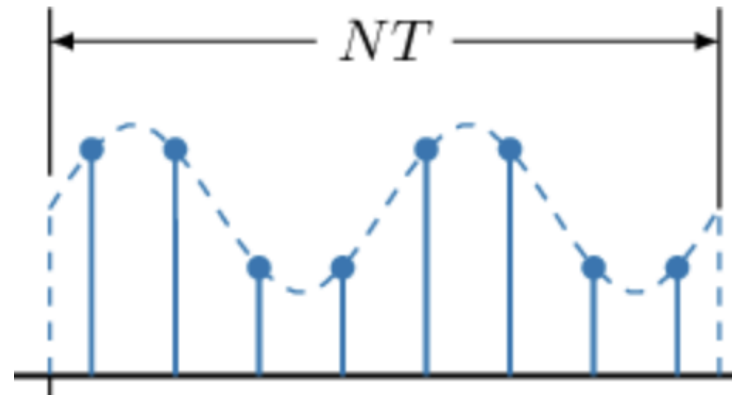


Discrete-Time Fourier Series (DTFS)



- Given a continuous signal $x(t)$ with period P : $x(t + P) = x(t)$
 - Fourier Series: $x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$, $\omega_0 = 2\pi/P$
 - Fourier coefficient: $c_k = \frac{1}{P} \int_0^P x(t) e^{-jk\omega_0 t} dt$
- After sampling: $x[n] = x[n + N]$
 - N samples within one period P , and the sampling period is T : $t = nT$ and $P = NT$

$$\omega_0 = \frac{2\pi}{P} = \frac{2\pi}{NT}$$



Discrete-Time Fourier Series (DTFS)

- A discrete signal $x[n]$ with period N : $x[n] = x[n + N]$ $\omega_0 = \frac{2\pi}{P} = \frac{2\pi}{NT}$
- N samples within one period P , and the sampling period is T : $t = nT$ and $P = NT$

- **Discrete-Time Fourier Series (DTFS):**

CTFS: $x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi \frac{k}{P} t} \xrightarrow{\text{Sampling...}} x[n] = \sum_{k=-\infty}^{\infty} \boxed{c_k} e^{\frac{j2\pi kn}{N}}$

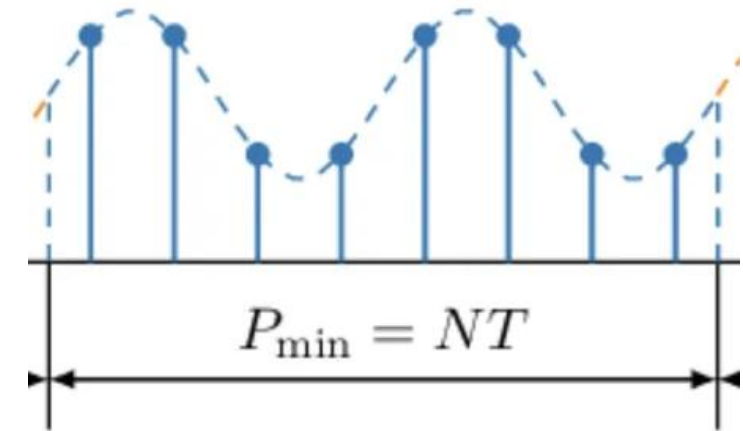
- Frequency basis becomes periodic!

$$e^{\frac{j2\pi(k+N)n}{N}} = e^{\frac{j2\pi kn}{N}}$$

" $k, k \pm N, k \pm 2N, \dots$ " becomes the same basis

DTFS: $x[n] = \sum_{k=0}^{N-1} \boxed{d_k} e^{\frac{j2\pi kn}{N}}$

How can we calculate d_k ?



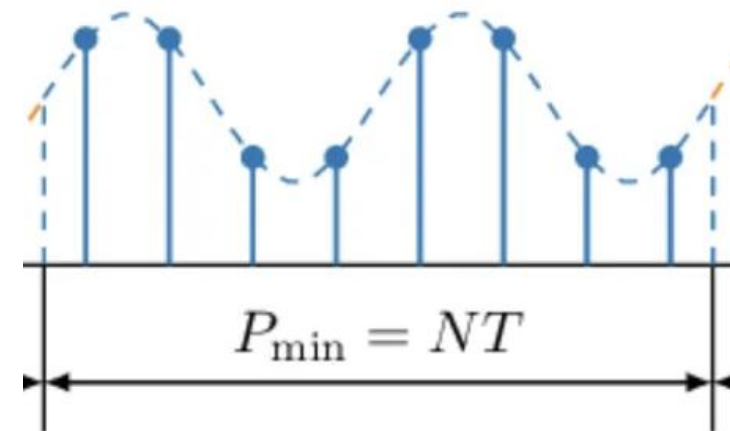
$$d_k = \sum_{r=-\infty}^{\infty} c_{k+rN} \rightarrow \text{Aliasing}$$

Discrete-Time Fourier Series (DTFS)

- A discrete signal $x[n]$ with period P : $x[n] = x[n + N]$ $\omega_0 = \frac{2\pi}{P} = \frac{2\pi}{NT}$
- N samples within one period P , and the sampling period is T : $t = nT$ and $P = NT$

- **DTFS coefficient:**

- Riemann sum: $\int_0^P f(t) dt \approx \sum_{n=0}^{N-1} f(nT)T$



CTFS coefficient $c_k = \frac{1}{P} \int_0^P x(t) e^{-jk\omega_0 t} dt \approx$

$$\frac{T}{P} \sum_{n=0}^{N-1} x(nT) e^{-j2\pi k(nT)/P} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} = d_k \quad \text{DTFS coefficient}$$

$$d_k = \sum_{r=-\infty}^{\infty} c_{k+rN} \xrightarrow{\text{Aliasing}} d_k - c_k = \sum_{r \neq 0} c_{k+rN}$$

Discrete-Time Fourier Series (DTFS)

- Aliasing!

DTFS
$$x[n] = \sum_{k=0}^{N-1} d_k e^{j2\pi kn/N}$$

DTFS coefficient
$$d_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

$$d_k = \sum_{r=-\infty}^{\infty} c_{k+rN}$$

$$d_k - c_k = \sum_{r \neq 0} c_{k+rN}$$

How can we overcome aliasing?

- Nyquist sampling condition!

It should become 0!

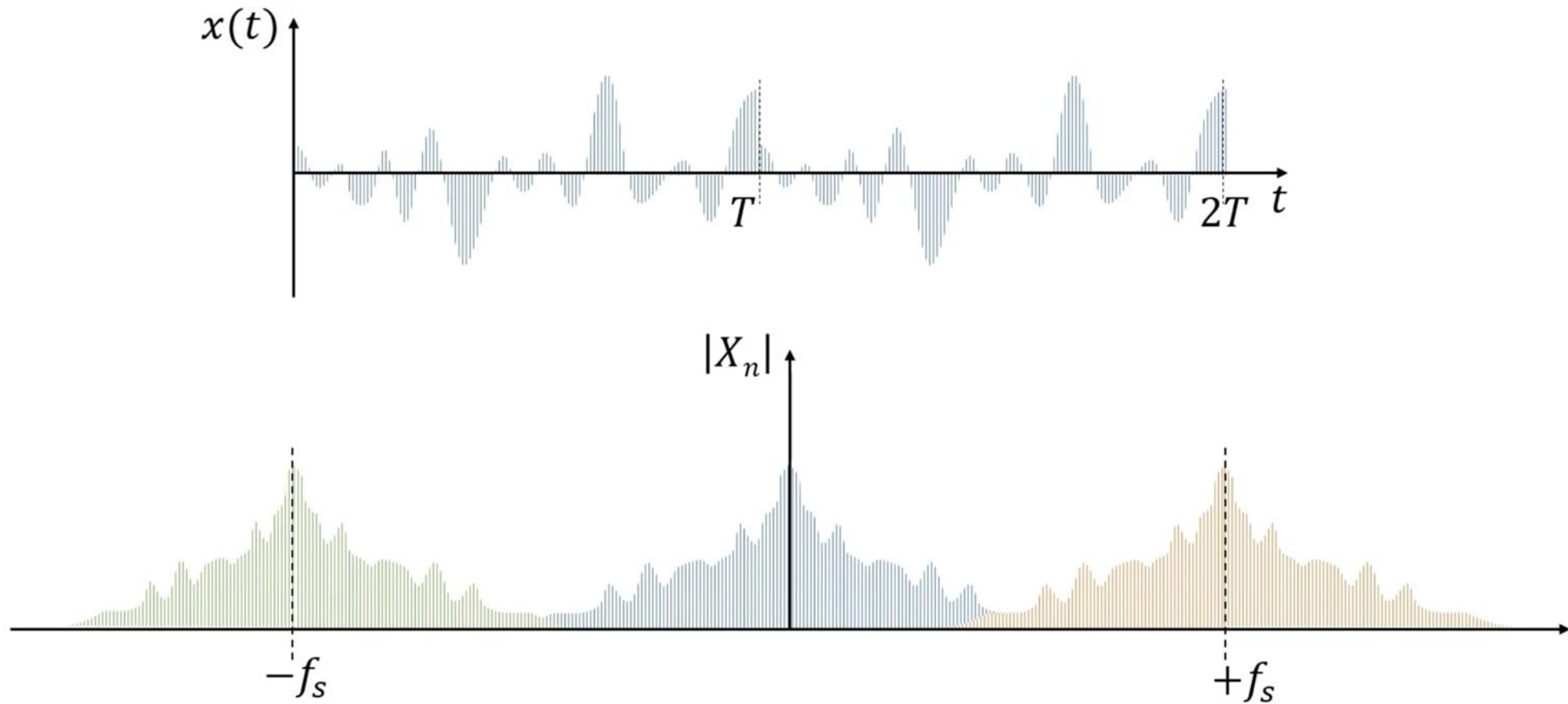
$$d_k - c_k = \sum_{r \neq 0} c_{k+rN}$$

Nonzero Fourier coefficient: $-K \leq k \leq K$

Zero Fourier coefficient ($c_k = 0$): $|k| > K$

- K: the highest harmonic (Highest nonzero harmonic index of $x(t)$)
- $f_{max} = \frac{K}{P}$ [Hz], $f_s = \frac{N}{P}$
- Frequency basis becomes periodic! $e^{\frac{j2\pi(k+N)n}{N}} = e^{\frac{j2\pi kn}{N}}$
 - " $k, k \pm N, k \pm 2N, \dots$ " becomes the same basis
- Unique frequency range
 - $-\frac{N}{2} < k < \frac{N}{2} \implies N > 2K \Leftrightarrow f_s > 2f_{max}$

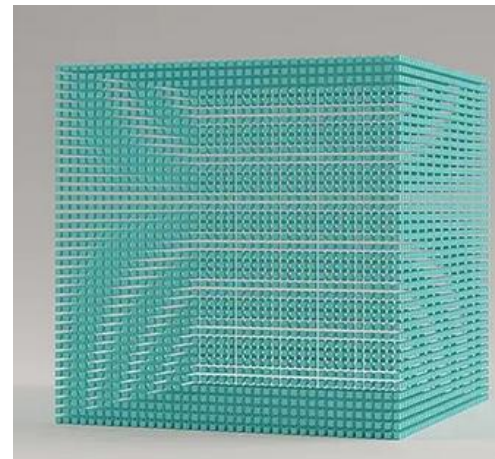
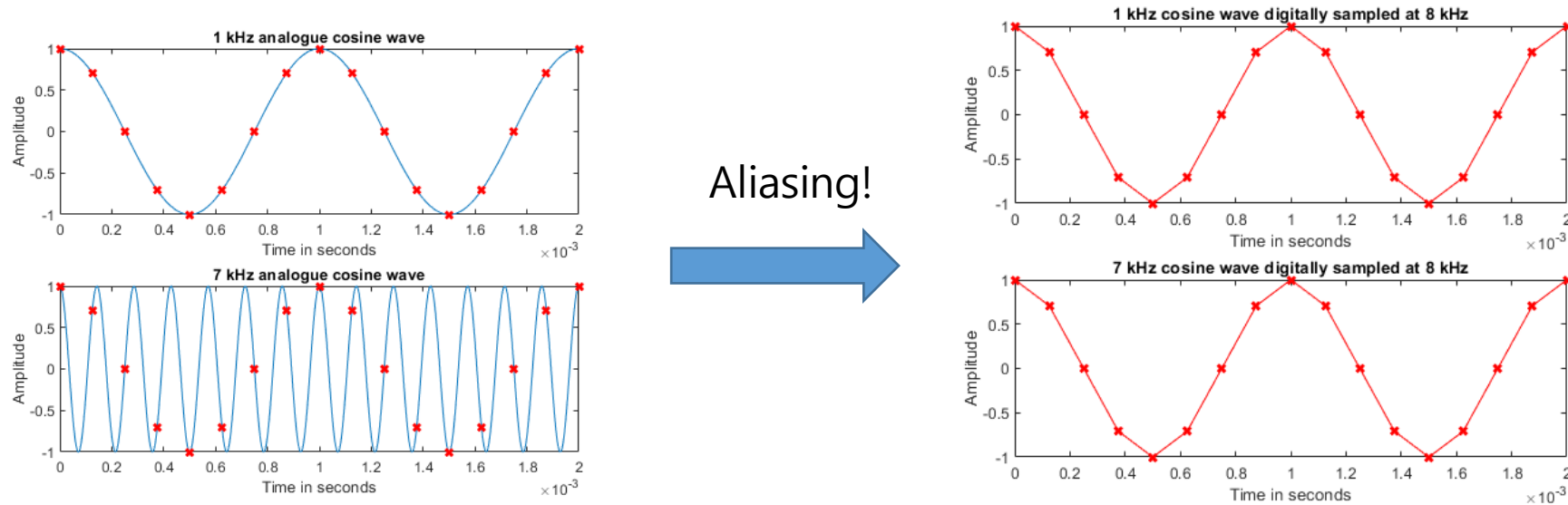
Discrete-Time Fourier Series (DTFS)



What if the Nyquist sampling condition is not satisfied?

Discrete-Time Fourier Series (DTFS)

- What if the Nyquist sampling condition is not satisfied?



Aliasing effect in images

https://en.wikipedia.org/wiki/Moir%C3%A9_pattern#
<https://www.adobe.com/creativelibrary/discover/anti-aliasing.html>

Discrete-Time Fourier Series (DTFS)

- DTFS with Nyquist sampling condition, $f_s > 2f_{max}$

DTFS
$$x[n] = \sum_{k=0}^{N-1} d_k e^{\frac{j2\pi kn}{N}}$$

$$d_k = c_k$$

DTFS coefficient
$$d_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

- What if the discrete-time signal is not periodic?
 - For example, suppose we have acquired N samples from a microphone.

$$x[0], x[1], \dots, x[N-1]$$

- There is no guarantee that this data is periodic, but we can regard the given N samples as one period.

$$\tilde{x}[n+N] = \tilde{x}[n] \quad \longrightarrow \quad \text{We can use the DTFS formula directly!}$$

Discrete Fourier Transform (DFT)

- Discrete Fourier Transform (DFT):

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N}n}$$

- Inverse Discrete Fourier Transform (IDFT):

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{n}{N}k}$$